José Carter Arriagada

Software Engineer | Real-Time XR, Physical AI & GPU Computing

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Profile

Software engineer (PUC Chile, 2025) building performance-critical software: real-time XR and AR research systems at DICTUC (full time) and physical-AI robotics software at TEK Chile (part-time freelance). Track record of measured speedups on GPUs and parallel systems: 15x in CUDA simulation and 10x in data pipelines.

Technical Skills

Languages: Python, TypeScript/JavaScript, C++, C#, Swift, Java, SQL **English:** Advanced (TOEIC 905)
GPU & Real-Time: CUDA, Metal, Unity, OpenXR, ARKit/RealityKit, eye and hand tracking, Isaac Sim, performance profiling
Robotics & Vision: YOLO, PyTorch, robot-arm and mobile-robot control, Modbus, TCP/IP, Raspberry Pi
Platforms: Node.js, React, Angular/Nx, Vue, Next.js, PostgreSQL, InfluxDB, MQTT, AWS, Docker, Railway, Linux, CI/CD

Experience

Multiplatform Full Stack Software Engineer, XR Lab Sep 2026 – Present
DICTUC - Pontificia Universidad Católica de Chile (full time) *Santiago, Chile*

- Moved a three-headset mixed-reality study app to a native **VIVE Focus Vision** build (**89.5 fps**, passthrough, hand and eye tracking) for ANID project 13250116 on emotion and collaboration in XR.
- Synchronized gaze, motion, and task events with **MindWare BioLab** physiological recordings to within a few milliseconds, with no turns lost on Wi-Fi drops; built a web panel that runs every headset from a PC or phone.
- Built Waves AR (Swift, **Metal FFT ocean**, ARKit) and a wireless dev loop that cut an AR iteration from **about 8 min to 4–7 s**.

Freelance Software Engineer, Physical AI & Robotics Sep 2026 – Present
TEK Chile (part time) *Santiago, Chile*

- Built the control app for a **Dobot CR5S** roll-packing cell (Raspberry Pi, tablet UI, TCP/IP and Modbus) where one plan drives both the 3D preview and the arm, plus an **Isaac Sim** digital twin and YOLO26 roll counting and box-corner vision.
- Built a browser controller with server-side safety for a **Dobot Rover X1** inspection robot: 18 real patrol runs with no contact, video raised from 5 to about 20 fps, Pi latency cut from **125 to about 25 ms**, and a 0.48 s emergency stop.

Full Stack Developer Sep 2025 – Sep 2026
Dily - Automotive Fintech Platform *Santiago, Chile*

- Built lending workflows (**TypeScript/Node.js**, Angular/Nx, PostgreSQL, AWS) with DDD, hexagonal architecture, and tests.

Lead Engineer, Alerting Platform Migration 2026
GridWorks *Santiago, Chile*

- Led the staged rebuild of an industrial alerting platform (Next.js, Node.js, MQTT, PostgreSQL) without interrupting live workflows.

Software Engineer Dec 2024 – Jul 2025
Flair - Commercial-Building Energy Optimization *Santiago, Chile*

- Python microservices, Vue.js UI, and InfluxDB time-series workflows on AWS, tied to **HVAC energy reductions of up to 50%**.

Research Software Engineer Jul 2023 – Jul 2025
Politiktok (Fondecyt) - Pontificia Universidad Católica de Chile *Santiago, Chile*

- GPU-backed Python/PostgreSQL pipelines for **100,000+ records**, about **10x faster** through parallelization; Springer co-author.

Simulation Engineer Jan 2023 – Jul 2024
Pontificia Universidad Católica de Chile - Geoscience *Santiago, Chile*

- Optimized **CUDA** parallel algorithms for tectonic-plate modeling and seismic-risk analysis, reaching a **15x speedup**.

AI Systems Developer Jan 2024 – Mar 2024
Drone Response - University of Notre Dame *Notre Dame, IN, USA*

- Mission planning for autonomous multi-drone coordination (Python, MQTT, OpenAI API, Java Spring); NASA/NSF-backed.

Teaching & Recognition

Teaching — Senior TA at PUC, 14+ roles in OS, HPC, Networks, XR (2023–2025); Python instructor, Escuela Militar (2024–2026).

Conferences — SC26 Early Career Program with travel grant (Chicago, 2026); ACM SIGGRAPH 2026 Student Volunteer.

Education

Pontificia Universidad Católica de Chile Mar 2020 – Jul 2025
B.S. Computer Science Engineering, Major in Software Engineering, Minor in Data Science; with Distinction *Santiago, Chile*